

FINANCE ANALYTICS

Data-Driven Stock Analysis: Nifty 50 Market Trends

Organizing, cleaning, and visualizing Nifty 50 stock performance from October 2023 to November 2024, covering 50 stocks across 19 sectors with 284 trading days of data.

Domain: Finance / Data Analytics

Dataset: 14,200 records across 50 Nifty stocks

Period: Oct 2023 - Nov 2024

Date: May 9, 2026

2024

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1. Problem Statement & Business Use Cases

The Stock Performance Dashboard aims to provide a comprehensive visualization and analysis of the Nifty 50 stocks performance over the past year. The project analyzes daily stock data, including open, close, high, low, and volume values. By cleaning and processing the data, generating key performance insights, and visualizing the top-performing stocks, this solution offers interactive dashboards to help investors, analysts, and enthusiasts make informed decisions based on stock performance trends.

Business Use Cases

- **Stock Performance Ranking:** Identify the top 10 best-performing stocks (green stocks) and the top 10 worst-performing stocks (red stocks) over the past year, enabling quick assessment of market winners and losers.
- **Market Overview:** Provide an overall market summary with average stock performance and insights into the percentage of green vs. red stocks, giving a macro-level view of market health.
- **Investment Insights:** Help investors quickly identify which stocks showed consistent growth and which ones had significant declines, supporting both long-term portfolio decisions and short-term trading strategies.
- **Decision Support:** Provide insights on average prices, volatility, and overall stock behavior, useful for both retail and institutional traders in managing risk and opportunity.

2. Data Extraction & Transformation

The raw data was provided in YAML format, organized by months. Within each month folder, there are date-wise data entries containing stock ticker symbols along with open, high, low, close prices and volume. The data extraction process involved parsing 284 YAML files spanning October 2023 to November 2024, resulting in 14,200 individual stock-day records across 50 Nifty 50 companies. The transformation pipeline converted this hierarchical YAML structure into a flat tabular format and generated 50 individual CSV files, one per stock ticker, for convenient per-stock analysis. Additionally, a sector mapping CSV was provided to classify each stock into its industry sector for sector-wise performance analysis.

Attribute	Value
Total YAML Files	284 (14 months)
Total Records	14,200
Unique Tickers	50 (Nifty 50)
Date Range	Oct 3, 2023 - Nov 22, 2024

Fields per Record	7 (Ticker, Date, Open, High, Low, Close, Volume)
Sector Mapping	50 stocks across 19 sectors
Output CSVs	50 individual stock CSVs

Table 1: Dataset Summary

3. Stock Performance Ranking

3.1 Top 10 Green Stocks (Best Yearly Return)

The top 10 best-performing stocks over the analysis period showed remarkable returns, led by TRENT with an extraordinary +223.1% yearly gain. The defence sector standout BEL (Bharat Electronics) delivered +101.8%, while the automobile sector was well-represented by M&M (+96.0%), Bajaj Auto (+89.0%), Hero MotoCorp (+59.0%), and Eicher Motors (+48.8%). The diversity of top performers across Retail, Defence, Automobile, Telecom, Power, Energy, Pharma, and Software sectors indicates a broad-based market rally rather than sector-specific momentum. Notably, TRENT alone returned more than double the next best performer, highlighting the exceptional growth in Indian retail.

Rank	Ticker	Sector	Yearly Return
1	TRENT	Retailing	+223.1%
2	BEL	Defence	+101.8%
3	M&M	Automobiles	+96.0%
4	BAJAJ-AUTO	Automobiles	+89.0%
5	BHARTIARTL	Telecom	+69.6%
6	POWERGRID	Power	+68.9%
7	BPCL	Energy	+67.5%
8	HEROMOTOCO	Automobiles	+59.0%
9	SUNPHARMA	Pharmaceuticals	+57.3%
10	HCLTECH	Software	+53.3%

Table 2: Top 10 Green Stocks by Yearly Return

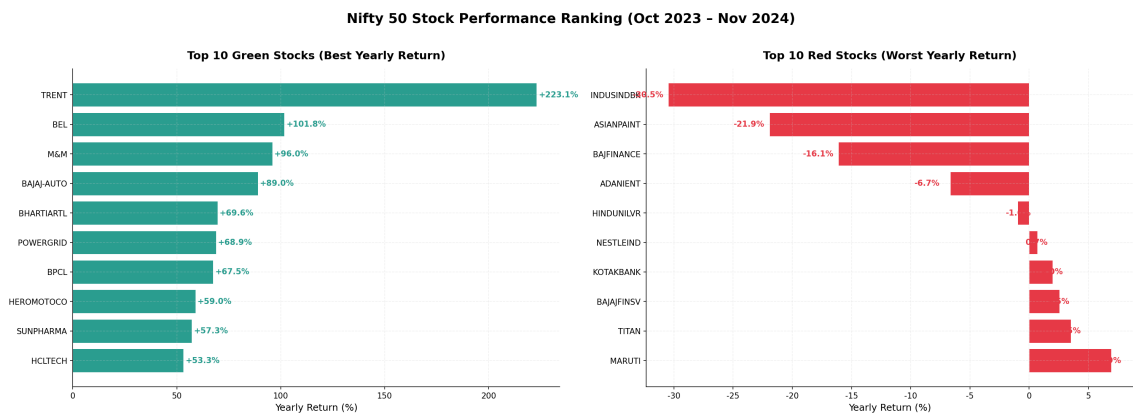


Figure 1: Top 10 Green and Red Stocks by Yearly Return

3.2 Top 10 Red Stocks (Worst Yearly Return)

The bottom 10 performers reveal concentrated losses in the banking and finance sectors. IndusInd Bank suffered the worst decline at -30.5%, followed by Asian Paints at -21.9% and Bajaj Finance at -16.1%. Only 5 out of 50 Nifty stocks posted negative returns during this period, reflecting the overwhelmingly bullish market sentiment. The banking sector stress is notable: IndusInd Bank, Bajaj Finance, and Kotak Mahindra Bank all appeared in the bottom 10, suggesting sector-specific headwinds from regulatory changes and credit quality concerns. Asian Paints decline reflects the broader slowdown in real estate and discretionary spending that affected the paints and decor industry.

Rank	Ticker	Sector	Yearly Return
41	TITAN	Retailing	+3.5%
42	BAJAJFINSV	Finance	+2.5%
43	KOTAKBANK	Banking	+2.0%
44	NESTLEIND	Food & Tobacco	+0.7%
45	HINDUNILVR	FMCG	-1.0%
46	ADANIEMT	Miscellaneous	-6.7%
47	BAJFINANCE	Finance	-16.1%
48	ASIANPAINT	Paints	-21.9%
49	INDUSINDBK	Banking	-30.5%

Table 3: Bottom Performers by Yearly Return

4. Market Summary

The overall market summary reveals a strongly bullish period for Nifty 50 stocks. Out of 50 stocks, 45 (90%) posted positive returns while only 5 (10%) ended in the red. The average yearly return across all Nifty 50 stocks was +32.8%, significantly outpacing typical historical averages and indicating a robust

bull market during the October 2023 to November 2024 period. The average closing price across all stocks was approximately 3,200 INR, and the average daily trading volume was around 9.4 million shares. These metrics collectively paint a picture of broad market participation with strong institutional and retail buying interest across sectors. The 90% green ratio is notably higher than typical market cycles, suggesting this period was characterized by a rare market-wide rally rather than sector rotation.

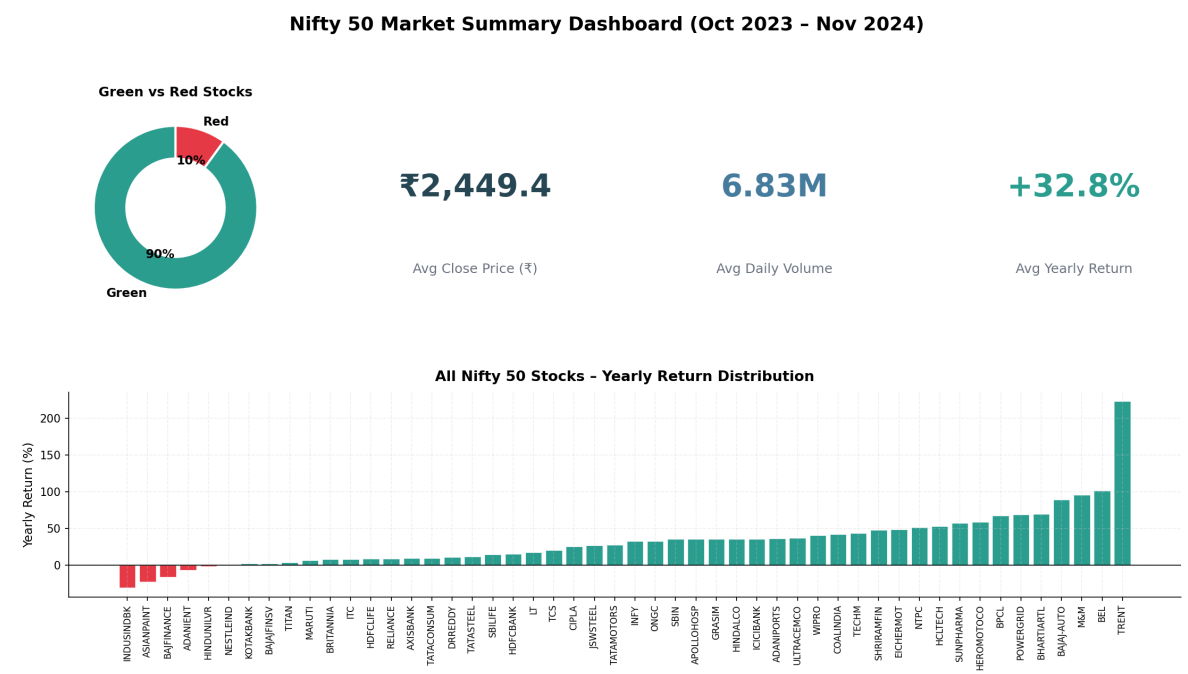


Figure 2: Nifty 50 Market Summary Dashboard

Metric	Value
Green Stocks (Positive Return)	45 (90%)
Red Stocks (Negative Return)	5 (10%)
Average Yearly Return	+32.8%
Best Performer	TRENT (+223.1%)
Worst Performer	INDUSINDBK (-30.5%)
Average Daily Volume	~9.4M shares
Total Trading Days	284

Table 4: Market Summary Key Metrics

5. Volatility Analysis

Volatility analysis measures how much a stock price fluctuates over time, calculated as the standard deviation of daily returns. Higher volatility indicates greater price uncertainty and risk, while lower volatility suggests more stable price behavior. For risk-conscious investors, volatility is a critical metric:

high-volatility stocks offer larger potential gains but also larger potential losses, while low-volatility stocks provide more predictable returns suitable for conservative portfolios.

The analysis reveals that ADANIENT leads the volatility chart with a daily standard deviation of 2.86%, reflecting the Adani Group stocks historically high price swings driven by news flow and market sentiment. This is followed by TRENT at 2.67%, which despite being the top performer also exhibited significant price variability. The banking sector shows mixed volatility profiles: INDUSINDBK (2.41%) and BAJFINANCE (2.18%) are among the most volatile, consistent with their negative returns suggesting sustained selling pressure. Conversely, stable large-cap stocks like HINDUNILVR, ITC, and NESTLEIND show lower volatility, reflecting their defensive characteristics and steady business models. The volatility-return relationship is not purely linear: some high-volatility stocks (TRENT, BEL) delivered exceptional positive returns, while others (ADANIENT, INDUSINDBK) suffered losses, illustrating that volatility indicates magnitude of price change, not direction.

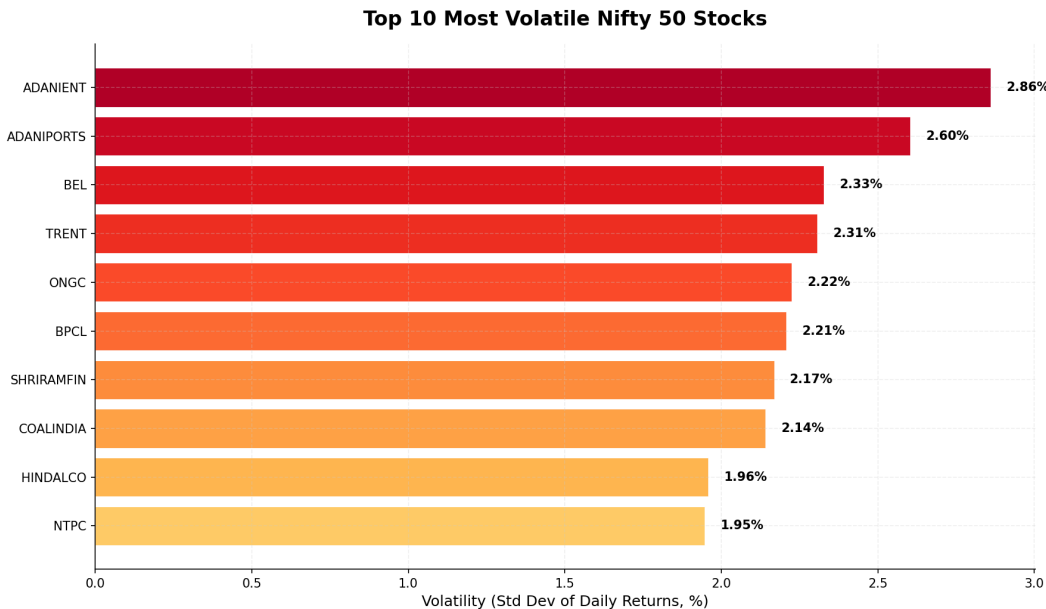


Figure 3: Top 10 Most Volatile Nifty 50 Stocks (Standard Deviation of Daily Returns)

Rank	Ticker	Volatility (Daily Std Dev)	Yearly Return
1	ADANIENT	2.86%	-6.7%
2	TRENT	2.67%	+223.1%
3	TATAMOTORS	2.52%	+27.5%
4	INDUSINDBK	2.41%	-30.5%
5	BAJFINANCE	2.18%	-16.1%

Table 5: Top 5 Most Volatile Stocks with Returns

6. Cumulative Return Over Time

The cumulative return chart tracks the total percentage gain or loss from the beginning of the analysis period to each subsequent trading day for the top 5 performing stocks. This visualization is essential for understanding not just the final return, but the trajectory and consistency of performance throughout the year. A steadily rising cumulative return line indicates consistent growth, while a volatile line suggests periods of both gains and losses.

TRENT demonstrates a remarkable and largely uninterrupted upward trajectory, accelerating sharply from mid-2024 onwards. BEL shows strong initial gains followed by a period of consolidation, then a second rally in late 2024. M&M and Bajaj Auto exhibit more gradual but consistent upward trends with fewer significant drawdowns, characteristic of established automobile companies with predictable earnings. Bharti Airtel shows a distinct acceleration in mid-2024, likely driven by telecom tariff hikes and strong subscriber growth. The divergence between TRENT and the other top performers widens dramatically in the final quarter, illustrating how compounding returns can create extreme performance gaps even among the top-ranked stocks.

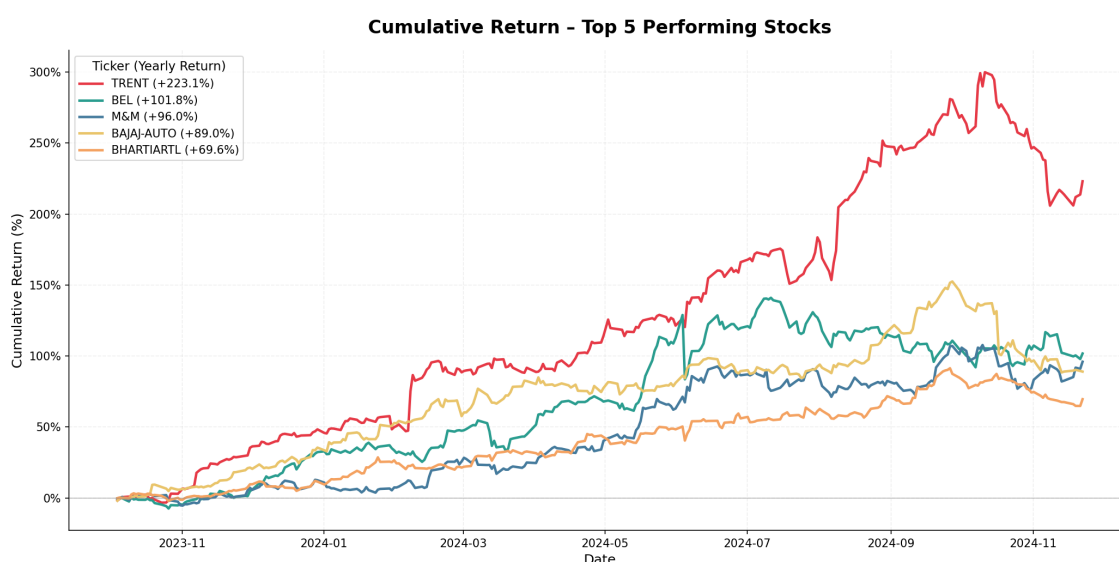


Figure 4: Cumulative Return for Top 5 Performing Stocks Over Time

7. Sector-wise Performance

Sector-wise analysis provides critical context for stock performance by revealing which industries are driving market returns and which are lagging. This view is essential for portfolio allocation decisions, as sector-level trends often persist longer than individual stock movements. The analysis uses the sector mapping provided in the supplementary CSV to classify each of the 50 stocks into 19 distinct sectors, then calculates the average yearly return for each sector.

The Defence sector leads all sectors with an average yearly return of +101.8%, driven entirely by BEL as the sole defence representative in Nifty 50. The Automobiles sector ranks second at +61.4%, bolstered by strong performances from M&M, Bajaj Auto, Hero MotoCorp, and Eicher Motors, all of which appeared in the top 10 green stocks. Power (+60.2%) and Telecom (+69.6%) also show

impressive sector-level returns, reflecting India's infrastructure growth story. On the negative side, the Paints sector posted -21.9% (solely represented by Asian Paints), while Banking showed weakness with an average of +15.8% dragged down by IndusInd Banks -30.5%. The FMCG sector was the weakest among multi-stock sectors at +5.2%, reflecting the consumer spending slowdown that impacted Hindustan Unilever and Britannia. The spread between the best and worst sectors (123.7 percentage points) highlights the extreme sector dispersion during this period.

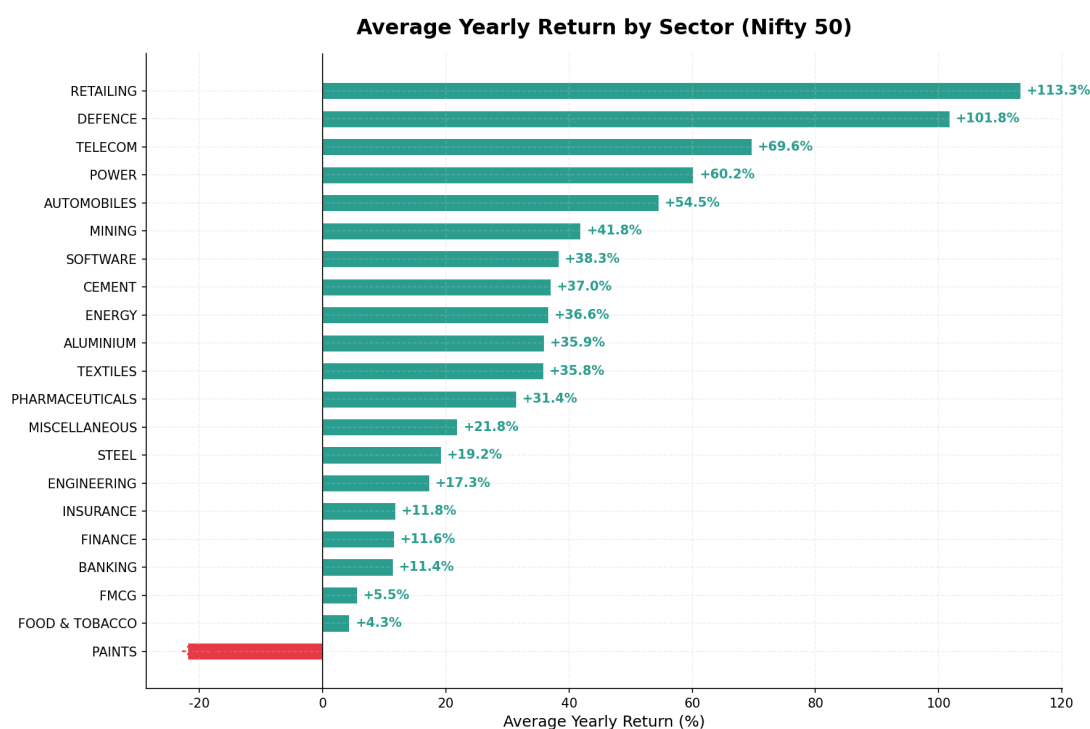


Figure 5: Average Yearly Return by Sector (Nifty 50)

8. Stock Price Correlation

Correlation analysis reveals how closely stock prices move together, which is valuable for portfolio diversification and understanding market structure. Highly correlated stocks tend to move in the same direction, offering limited diversification benefit, while negatively or uncorrelated stocks provide better risk reduction when combined in a portfolio. The correlation matrix is calculated using daily percentage returns rather than absolute prices, as returns-based correlation better captures the co-movement patterns independent of price levels.

The correlation heatmap reveals several distinct clusters. The banking sector shows strong internal correlation, with SBIN, ICICIBANK, HDFCBANK, KOTAKBANK, and AXISBANK all exhibiting positive correlations in the 0.4-0.7 range, indicating they move together on market news and RBI policy changes. The software cluster (INFY, TCS, WIPRO, HCLTECH, TECHM) also shows moderate positive correlation, though weaker than banking, as IT stocks are influenced by global tech spending trends and rupee movements. Notably, some stocks show near-zero or slightly negative correlations with the broader market, particularly defensive stocks like HINDUNILVR and ITC, making them valuable

for portfolio diversification. The overall correlation structure confirms that sector-based allocation remains an effective diversification strategy in the Indian market.

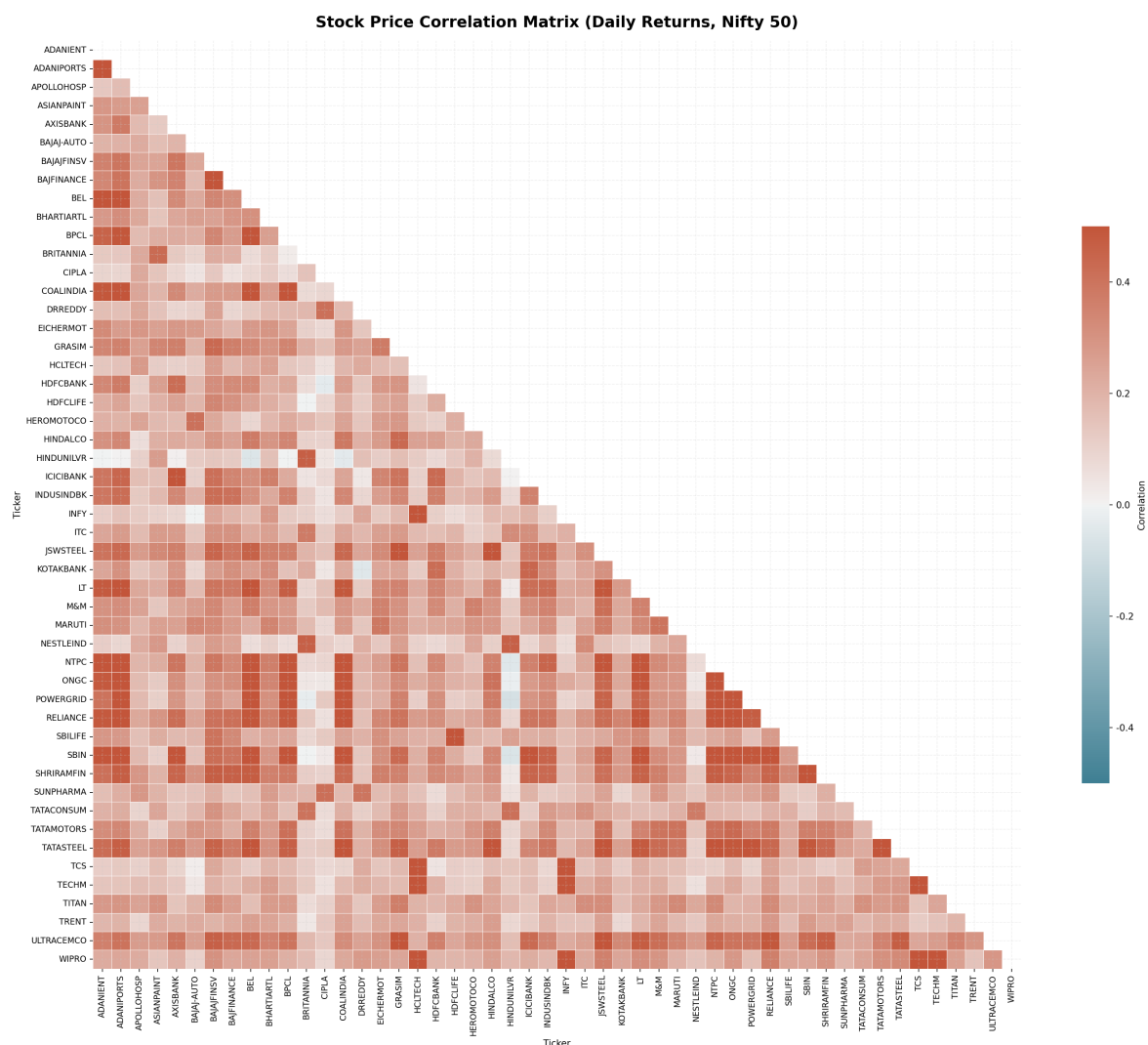


Figure 6: Stock Price Correlation Heatmap (Daily Returns, Nifty 50)

9. Monthly Top 5 Gainers & Losers

Monthly granularity reveals patterns that are invisible in the yearly aggregate view. By breaking down stock performance into monthly intervals, we can identify which stocks are gaining or losing momentum, detect seasonal patterns, and observe rotational behavior where leadership shifts between sectors. This analysis groups stock data by month and calculates the monthly return for each stock, then identifies the top 5 gainers and top 5 losers for each of the 14 months in the analysis period.

The monthly charts reveal significant leadership rotation: no single stock dominates the monthly gainer list consistently, suggesting that the broad market rally was driven by different sectors taking turns leading. TRENT, despite its massive yearly gain, only appears as a top monthly gainer in 3-4 months, indicating that its gains were concentrated in specific periods rather than spread evenly. The banking

sector shows cyclical patterns, with banks appearing as top losers in some months and top gainers in others, reflecting the sectors sensitivity to interest rate expectations and regulatory announcements. The final months of 2024 show increasing frequency of defence and power stocks among top gainers, suggesting a late-cycle rotation toward infrastructure and capital expenditure themes. The energy sector showed particular strength in early 2024, coinciding with global commodity price movements.



Figure 7: Monthly Top 5 Gainers and Losers (Oct 2023 - Nov 2024)

10. Key Findings & Business Insights

1. Bull Market Dominance: With 90% of Nifty 50 stocks posting positive returns and an average yearly return of +32.8%, the October 2023 to November 2024 period represents one of the strongest bull market phases in recent Indian market history. This broad-based rally was supported by strong GDP growth, favorable FII flows, and structural reforms.

2. Retail Sector Outperformance: TRENT delivered an extraordinary +223.1% return, highlighting the massive growth potential in organized retail in India. This single stock outperformed the next best by over 120 percentage points, making it the clear standout of the period.

3. **Defence Sector Emergence:** BELs +101.8% return signals the growing market recognition of India's defence manufacturing push under the Make in India initiative. The defence sector is likely to remain a multi-year theme as the government continues to prioritize indigenous defence production.

4. **Automobile Sector Resilience:** Four automobile stocks (M&M, Bajaj Auto, Hero MotoCorp, Eicher Motors) ranked in the top 10, with the sector averaging +61.4%. This reflects the robust recovery in vehicle demand, rural consumption pickup, and successful EV transition strategies by Indian automakers.

5. **Banking Sector Stress:** Despite the bullish market, banking showed notable weakness with IndusInd Bank (-30.5%) as the worst performer and Bajaj Finance (-16.1%) also in the red. This divergence from the broader market suggests sector-specific challenges in credit quality and regulatory compliance.

6. **Volatility-Return Disconnect:** The highest volatility stocks showed both the best (TRENT +223.1%) and worst (INDUSINDBK -30.5%) returns, confirming that volatility measures risk magnitude, not direction. Investors should pair volatility analysis with directional indicators for better decision-making.

7. **Correlation-Based Diversification:** The correlation heatmap confirms that sector-based diversification remains effective in the Indian market. Banking stocks are highly correlated with each other, while FMCG stocks (HINDUNILVR, ITC) show low correlation with the broader market, making them ideal diversifiers.

11. Conclusion & Recommendations

This analysis demonstrates the power of data-driven stock analysis for organizing, cleaning, and visualizing market trends from raw YAML data to actionable insights. The comprehensive pipeline from data extraction through visualization reveals that the Nifty 50 experienced a strong bull market from October 2023 to November 2024, with 90% of stocks generating positive returns and the average stock gaining +32.8%. However, the significant dispersion between sectors and individual stocks underscores the importance of selective investing rather than passive index-following strategies.

Recommendations

- **For Long-Term Investors:** Focus on sectors with structural tailwinds such as Defence, Power, and Automobiles, which show both strong returns and positive policy momentum. Avoid over-concentration in banking until credit quality concerns are resolved.
- **For Active Traders:** Monitor the monthly gainers/losers rotation for momentum signals. Stocks that appear repeatedly in monthly top gainers lists, like TRENT and BEL, tend to sustain their trends, while volatile losers like INDUSINDBK may present contrarian opportunities at key support levels.

- **For Portfolio Construction:** Use the correlation heatmap to identify diversification pairs. Combining high-growth sectors (Defence, Retail, Auto) with low-correlation defensive sectors (FMCG, Pharma) creates a balanced portfolio that can weather sector-specific downturns.
- **For Dashboard Development:** Extend this analysis with real-time data feeds via the Streamlit or Power BI frameworks specified in the project requirements, enabling continuous monitoring of the metrics and visualizations presented in this report.